INVERTED MAML RESEARCH TEAM — MULTI-AGENT ARCHITECTURE

Autonomous Multi-Agent Organization, Responsibilities, and Adversarial Scientific Protocol

Patent Pending: SE 2630397-4 | **Hardware:** Trion T120 SoC + T20 Coprocessor | **Date:** 2026-10-16 21:29:39 CEST

The Inverted MAML research framework operates as a specialized multi-agent hierarchy designed to decouple strategic direction, physical hardware execution, mathematical modeling, and rigorous adversarial validation. This structure guarantees reproducible research, prevents confirmation bias, strictly enforces hardware safety (heaters disarmed, storage budgets < 100 KB), and ensures that physical jumper configurations (Modes A, B, and C) are respected at all times.

1. Multi-Agent Functional Block Architecture

PILLAR 1: STRATEGIC LEADERSHIP	PILLAR 2: HARDWARE & EMULATION	PILLAR 3: ALGORITHM & DATA	PILLAR 4: ANTI-SIMULATION & AUDIT
maml_team_leader Overall picture, roadmap orchestration, and datacenter perspective. **maml_public_rep** Public liaison, human reporting, and PDF dossier generation. **maml_research_planner** Firmware changelog maintainer, scheduling, and gatekeeper.	**maml_hardware_expert** *Primary Custodian for Jumper Modes A/B/C.* Enforces no-training rule in Mode B. Manages heater lockout (HEAT=0). **maml_hardware_emulator_expert** Physical Hidden-Offset Emulation (Patent Fig. 7: W + Hidden_offset, DAC(2), trim-DAC 38) on real PCB.	**maml_nn_specialist** Inverted MAML outer loop 40, inner loop 41, and 2x6 Hadamard (24 us) gradient extraction. **maml_data_crawler** UDP binary telemetry (0x4D414D4C), live hardware captures. **maml_data_analyzer** Computes real SNR, derivatives, and ADC response metrics.	**maml_anti_simulation_enforcer** *ZERO-SIMULATION WATCHDOG:* Strictly bans all software simulations and synthetic surrogates. **maml_bug_methodology_critic** *'The Devil's Advocate'*: Challenges oscilloscope signals, op-amp drive limits, and ADC settling.

2. Operational Responsibilities & Inter-Agent Protocols

Agent Name	Primary Mandate	Key Deliverables & Enforced Constraints
maml_anti_simulation_enforcer	**Zero-Simulation Watchdog:** Total ban on all software simulations in measurements and NN training.	**FORBIDS all synthetic surrogates.** Only real physical ADC measurements and real hardware hidden-offset emulation (Fig. 7) allowed.
maml_team_leader	Synthesizes overall research trajectory, balances model architecture with physical silicon limits.	Ensures hardware-first verification and strict scientific honesty.
maml_hardware_expert	Hardware Custodian: Jumper Modes (A/B/C), analog multi-carrier buses, and thermal safety.	**Enforces Jumper Mode B strict no-training rule.** Interlocks HEAT_1..4_EN to 0% duty at all times.
maml_hardware_emulator_expert	Physical Hardware Hidden-Offset Emulation (Patent Fig. 7 / Ref 38, 40, 41) on real DACs.	Applies programmatic hidden offsets to physical 74HC595 DACs and trim-DAC 38. Emulation != Simulation.
maml_nn_specialist	Oversees non-backpropagation Hadamard gradient descent (Eqs. 1–7) on physical hardware.	Implements contiguous 2x6 (12 x 2 us = 24 us) Hadamard bursts with zero tri-state gaps between pulses.
maml_bug_methodology_critic	Adversarial Skeptic: Audits oscilloscope traces, op-amp current drive, and ADC timing.	Verifies that ADC sampling occurs strictly at the end of each 2 us pulse and within the flat post-refresh window.
maml_research_planner	Maintains system changelog, software builds, and execution sequence.	Authors firmware_changelog.md with exact release dates, bitstream hashes, and hardware verification steps.

3. Core Operational Directives

• **1. Total Ban on Simulations:** Zero software simulations in measurements or NN training. Only physical hardware and real hidden weights (Patent Fig. 7) are permitted.

• **2. Contiguous 24 us Hadamard Table:** 2x6 pulses (6 (+) and 6 (-)), each exactly 2 us long with ADC sampling at the end of each 2 us pulse and zero tri-state gaps between pulses.

• **3. Zero Unsupervised Thermal Risk:** Power heaters (HEAT_1..4_EN) remain permanently disarmed (duty = 0) to protect hardware.

• **4. Jumper Discipline:** Mode B (bare silicon) is purely diagnostic; compensation weights are frozen to prevent divergence.